EE4-24
EE9-CS1-3

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

MSc and EEE/EIE PART IV: MEng and ACGI

DIGITAL CONTROL SYSTEMS

Friday, 7 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : A. Astolfi

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DIGITAL CONTROL SYSTEMS

Information for candidates:

- $Z\left(\frac{1}{s}\right) = Z(1(kT)) = \frac{z}{z-1}$
- $Z\left(\frac{1}{s+a}\right) = Z\left(e^{-akT}\right) = \frac{z}{z-e^{-aT}}$
- $Z\left(\frac{\omega}{s^2+\omega^2}\right) = Z(\sin \omega kT) = \frac{z \sin \omega T}{z^2 - 2z\cos \omega T + 1}$
- $Z\left(\frac{1}{s^2}\right) = Z(kT) = \frac{zT}{(z-1)^2}$
- Let $F(s)$ be the Laplace transform of $f(t)$ and n a positive integer. Then

$$Z\left(e^{-nTs}F(s)\right) = Z(f(kT-nT)1(kT-nT)) = z^{-n}Z(f(kT))$$

- Transfer function of the ZOH: $H_0(s) = \frac{1-e^{-sT}}{s}$
- Definition of the w-plane: $z = \frac{1+\frac{wT}{2}}{1-\frac{wT}{2}}$, $w = \frac{2}{T}\frac{z-1}{z+1}$
- Note that, for a given signal r (also indicated as $r(k)$, $r(kT)$ or $r(t)$), $R(z)$ denotes its Z-transform. All signals are assumed to be zero for negative times.

1. a) For each of the following items (from i to viii), sketch the time-sequence associated to the pole locations in the z-plane. Assume that all poles are simple and that the time-sequence is the response to a unit impulse. Sketch at least 9 samples. For instance, the time-sequence of a pole at $z = 1$ is shown in the figure below. [8 marks]

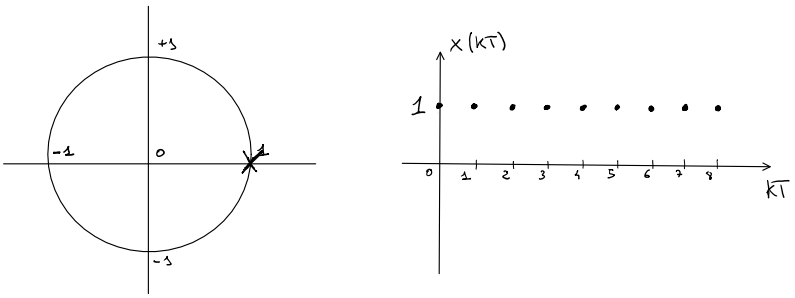
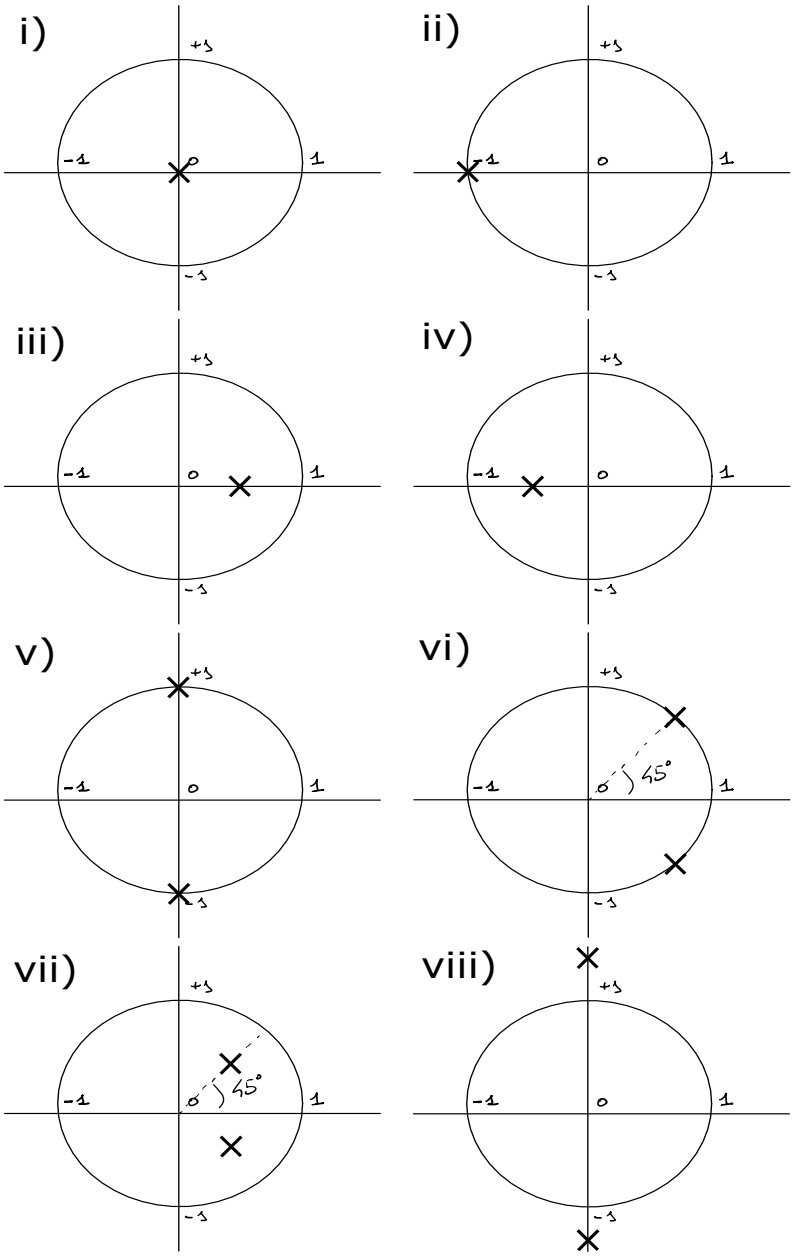


Figure 1.1 Example of impulsive response for one pole at $z = 1$.



[8 marks]

b) Consider the continuous-time system shown in Figure 1.2, where

$$G(s) = \frac{s^2}{s^2+4},$$

and its discretized version shown in Figure 2.1, where $H_0(s)$ is a zero-order hold. Let $T > 0$ be the sampling time.

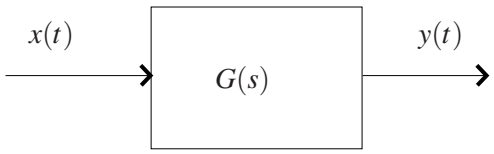


Figure 1.2 Continuous-time system for question 1.

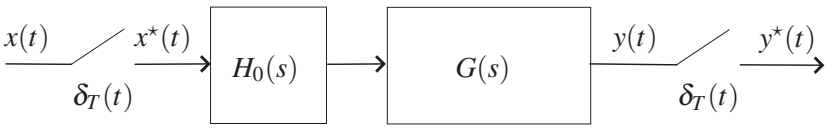


Figure 1.3 Discretized version of the system.

i) Discuss the stability properties of the continuous-time system shown in Figure 1.2. Let $x(t)$ be the unit step. Compute $Y(s)$ and the corresponding continuous-time signal $y(t)$.

[2 marks]

ii) Without using any computation, discuss the stability properties of the discrete-time system shown in Figure 2.1 (justify your reasoning). Let $x(t)$ be the unit step. Compute $Y(z)$.

[3 marks]

iii) Set $T = \frac{\pi}{4}$. Use the direct division method to write $Y(z)$ (obtained in part b.ii)) as an infinite series and then write the first 5 samples of the sequence $y(kT)$, that is $y(0)$, $y(T)$, $y(2T)$, $y(3T)$ and $y(4T)$.

[2 marks]

iv) Set $T = \pi$. Compute (with any method you prefer) the inverse z-transform of $Y(z)$ (obtained in part b.ii)) and then write the first 5 samples of the sequence $y(kT)$, that is $y(0)$, $y(T)$, $y(2T)$, $y(3T)$ and $y(4T)$.

[2 marks]

v) On the basis of the samples of the sequences computed in parts b.iii) and b.iv), can you assess whether the continuous-time signal $y(t)$ is constant or oscillating? Justify your answer and explain what phenomenon you are observing.

[3 marks]

2. Consider the block diagram in Figure 2.1, where $H_{\triangle}(s)$ represents the transfer function of a so-called triangle-hold.

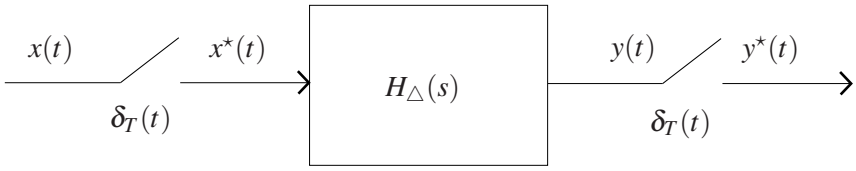
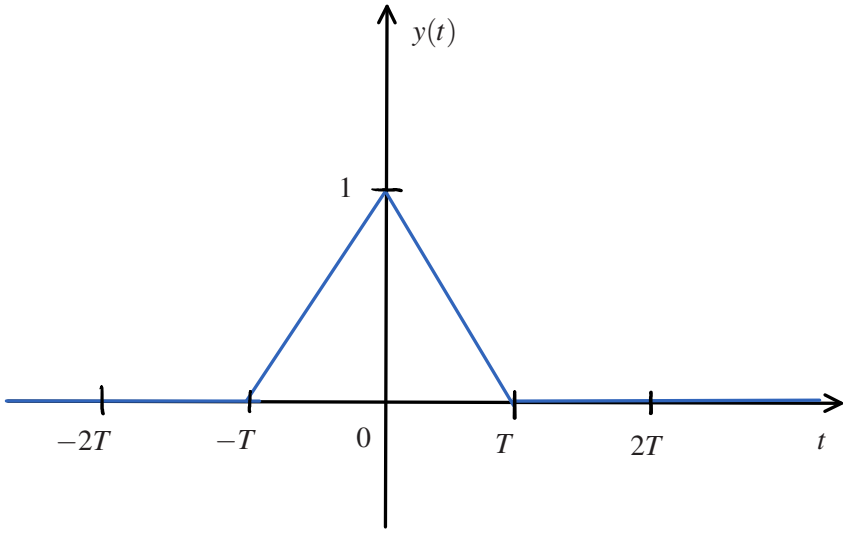


Figure 2.1 Triangle-hold interconnection.

Let $T > 0$ be the sampling time. The figure below shows the output $y(t)$ of a triangle-hold when the input is a unit impulse.



a) Determine the transfer function $H_{\triangle}(s)$ of the triangle-hold. To this end, follow these steps.

i) Write the equation of the output $y(t)$ of the hold as a function of the input $x(kT)$ (recall that x is the unit impulse). [4 marks]

ii) Find $Y(s)$, $X^*(s)$ (*i.e.* the starred Laplace transform of the input) and determine the transfer function $H_{\triangle}(s)$ of the triangle-hold using $Y(s)$ and $X^*(s)$. [4 marks]

b) Let $G(s)$ be any transfer function. Show that $Z[H_{\triangle}(s)G(s)]$ (namely, the z-transform of the interconnection of $G(s)$ with a triangle-hold) can be written as $f(z)Z\left[\frac{G(s)}{s^2}\right]$, where $f(z)$ is a function of z. Determine $f(z)$. [4 marks]

c) Prove that the frequency response of the triangle-hold can be written as

$$H_{\triangle}(j\omega) = T\left(\frac{\sin\left(\frac{T\omega}{2}\right)}{\frac{T\omega}{2}}\right)^2.$$

[4 marks]

d) Compare the delays of the zero-order hold and the first-order hold with the delay of the triangle-hold. Is the triangle-hold an acceptable choice in practical control scenarios? [4 marks]

3. Consider the digital control system in Figure 3.1

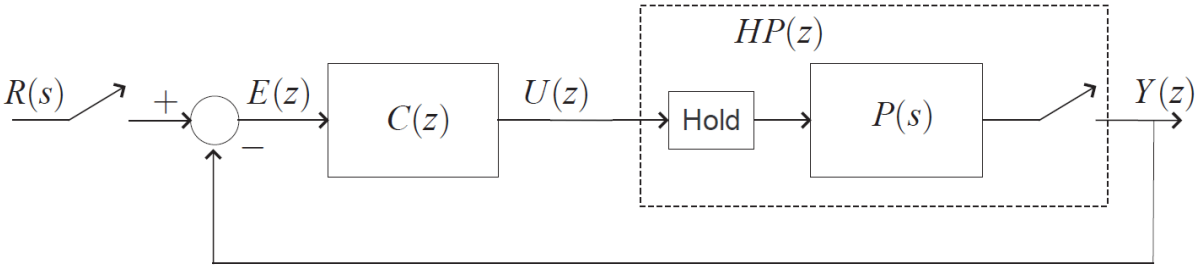


Figure 3.1 Block diagram for question 3.

with transfer function

$$P(s) = \frac{1}{s(s+a)}.$$

Let $T > 0$ be the sampling time.

a) Compute the equivalent discrete-time model $HP(z)$ of the plant interconnected to a zero-order hold and a sampler. Show all the steps of your work, in particular, show the partial fraction expansion, the time domain signals obtained using the inverse Laplace transform and, finally, the expression in the z-domain. [5 marks]

b) For $a = 2$ and $T = \ln(2)$, the equivalent transfer function in the w-plane is given by

$$HP(w) = \frac{3-5\ln(2)}{20} \frac{\left(w + \frac{6}{5\ln(2)-3}\right)\left(w - \frac{2}{\ln(2)}\right)}{w^2 + \frac{6}{5\ln(2)}w}.$$

Apart for part e) below, do not approximate $\ln(2)$ with its decimal representation. Consider a controller

$$C(w) = \frac{\alpha}{w+\beta}.$$

For two values of β there is a cancellation in the product $C(w)HP(w)$. Pick β such that there is a cancellation and explain why you have selected that specific value instead of the other. [4 marks]

c) Study the stability properties of the closed-loop system as a function of α. [4 marks]

d) Determine all the values of α such that the closed-loop system is asymptotically stable and the position velocity constant is at least 1. [4 marks]

e) Using the result of part d), select a value of α such that the closed-loop system is asymptotically stable and the position velocity constant is at least 1. Use the approximation $\ln(2) \approx 0.6931$ and compute the equivalent discrete-time controller $C(z)$. Does this discrete-time controller stabilize the discrete-time closed-loop systems and achieve a velocity constant of at least 1? Justify your answer. [3 marks]

4. Consider the digital control system in Figure 4.1.

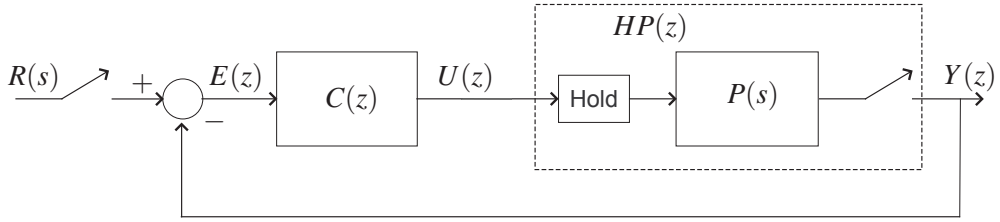


Figure 4.1 Block diagram for question 4.

The continuous-time system is defined by

$$\begin{aligned} \dot{x}(t) &= Fx(t)+Gu(t), \\ y(t) &= Hx(t), \end{aligned} \tag{4.1}$$

with

$$F=\begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix}, \qquad G=\begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad H=\begin{bmatrix} 1 & 0 \end{bmatrix}.$$

Let $T>0$ be the sampling time.

a) Discretize the continuous-time state-space equations (4.1). To this end, determine the matrices A, B and C of the representation

$$\begin{aligned} x(k+1) &= Ax(k)+Bu(k), \\ y(k) &= Cx(k), \end{aligned} \tag{4.2}$$

when the input $u(t)$ is sampled and fed to a zero-order hold.
Hint: recall that $e^{Ft}=\mathscr{L}^{-1}\left[(sI-F)^{-1}\right]$, *where* $\mathscr{L}^{-1}$ *indicates the inverse Laplace transform operator.*

[4 marks]

b) Compute the input-output transfer function of the discrete-time system (4.2).

[3 marks]

c) Without computing the reachability and observability matrices, assess whether system (4.2) is reachable and/or observable. Explain your reasoning.

[2 marks]

d) In practical scenarios it is often impossible to access all the components of the state of a system. In these cases we need to reconstruct the state by using an estimator which uses measurements of the output. Consider the following estimator

$$\tilde{x}(k+1)=A\tilde{x}(k)+Bu(k)+L[y(k)-C\tilde{x}(k)], \tag{4.3}$$

where $\tilde{x}(k)$ is an estimate of $x(k)$. Show that if the eigenvalues of the matrix $A-LC$ are all inside the unit circle, then the estimation error $e(k)=\tilde{x}(k)-x(k)\to 0$ as $k\to\infty$.

[3 marks]

e) Let $T=0.1$. Find L such that eigenvalues of the matrix $A-LC$ are $0.4\pm j0.4$.

[4 marks]

f) Consider the control $u(k)=-K\tilde{x}(k)$, where $\tilde{x}(k)$ is given by (4.3). Determine conditions on K and L such that $e(k)\to 0$ and $x(k)\to 0$ as $k\to\infty$.

[4 marks]